Diagnose, repair, recheck

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify a reasoning error and demonstrate the repaired method on a new problem.

SUCCESS CRITERIA - I can...

1. Locate the first incorrect step.
2. Show a corrected method with units.
3. Apply the repair to a different problem.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 kg cart moves left at 5 m/s. Forces are 26 N right and 5 N left. Find acceleration and whether its speed initially rises or falls.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a = \Sigma F/m$; $v_f = v_i + a\,\Delta t$.
For a constant-acceleration interval. Check signs, units and assumptions; keep the original step visible when revising.

Worked reasoning: Claim: a car moving east must accelerate east. Counterexample: it moves east while braking, so acceleration points west. A single valid counterexample can disprove a universal claim.

Your turn: Draw velocity and acceleration arrows for that counterexample. Write a corrected statement about their directions.

Car braking while moving east

East is right.

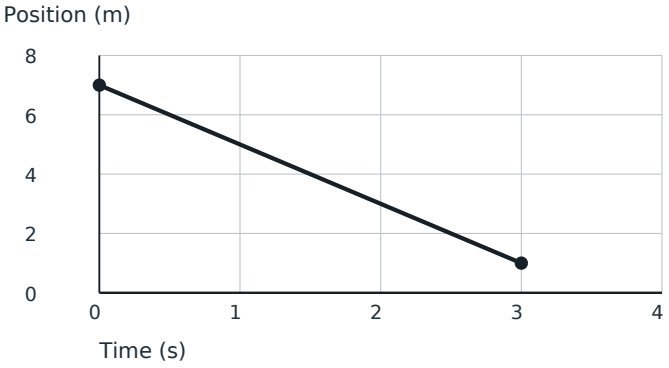
Velocity:

Acceleration / force:

Marzano check - circle: ① ② ③

03 Position is not velocity

A cart's position changes from 7 m to 1 m in 3 s, as shown. A student says, "Its velocity is 1 m/s because it ends at 1 m." Explain the mistake and calculate the velocity correctly.



Marzano check - circle: ① ② ③

04 Net force is not one force

A 4 kg cart has 18 N right and 6 N left. A student calculates acceleration as 18 ÷ 4. Use the force diagram to explain the missing step, then correct the calculation.



Marzano check - circle: ① ② ③

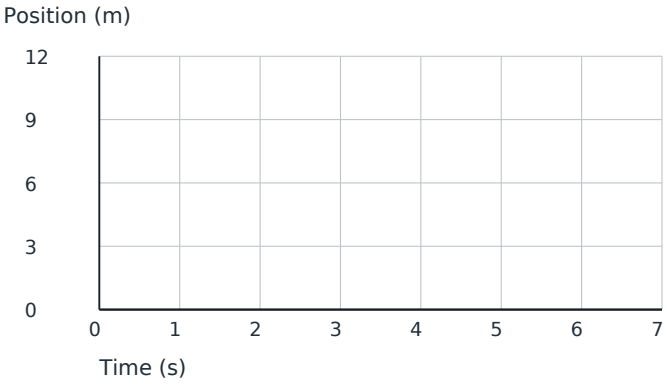
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Continue down each column.

05 New evidence

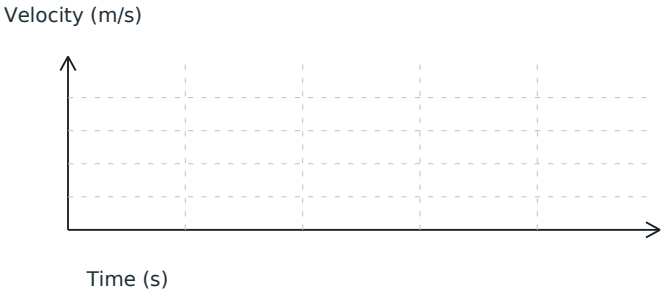
Try two new checks.
(a) A robot moves steadily from 10 m to 2 m in 7 s. Find its velocity.
(b) A 5 kg cart has 7 N right and 22 N left. Find its acceleration. Show how each calculation accounts for direction.



Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart travels west at 14 m/s and west at 5 m/s 4 s later. Is it speeding up? Find acceleration size and direction. Sketch the velocity-time model; label the given values.



Marzano check - circle: 1 2 3

07 Exit check

Choose the error in question 3 or 4. Explain the correction, then point to the matching new check in question 5. What can you now do without help?

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Invent a wrong statement about force or motion, then draw a specific case that disproves it. State the corrected rule.

Add and label force arrows.

Use a diagram to test the statement you invent.

Marzano check - circle: 1 2 3